jupyter Assignment No. 6 Last Checkpoint: 2 days ago

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Code JupyterLab Python (ml_env)

[1]:
```
print("=" * 65)
print("        BASIC DATA PREPROCESSING USING PYTHON")
print("                  UCI IRIS DATASET")
print("=" * 65)

print("\nTasks Performed:")
print("1. Dataset Loading")
print("2. Dataset Exploration")
print("3. Missing Value Handling")
print("4. Categorical Data Encoding")
print("5. Numerical Data Normalization")
print("6. Final Preprocessed Dataset")
print("=" * 65)
```

```
=================================================================
        BASIC DATA PREPROCESSING USING PYTHON
                  UCI IRIS DATASET
=================================================================

Tasks Performed:
1. Dataset Loading
2. Dataset Exploration
3. Missing Value Handling
4. Categorical Data Encoding
5. Numerical Data Normalization
6. Final Preprocessed Dataset
=================================================================
```

[2]:
```
import pandas as pd
import numpy as np

from sklearn.preprocessing import LabelEncoder, MinMaxScaler

print("✓ Required libraries imported successfully.")
```

```
✓ Required libraries imported successfully.
```

[ ]:

[3]:
```
import os

items = os.listdir()

print("Files and folders in the current directory:\n")

for item in items:
    print("📁" if os.path.isdir(item) else "📄", item)
```

```
Files and folders in the current directory:

📁 .ipynb_checkpoints
📄 Assignment No. 6.ipynb
📄 Experiment4.ipynb
📄 Experiment5.ipynb
📄 Untitled.ipynb
```

[7]:
```
import os

dataset_folder = r"D:\AIML LAB"

print("Dataset folder:")
print(dataset_folder)

print("\nFiles available:")
print("-" * 50)

for item in os.listdir(dataset_folder):
    print("📁" if os.path.isdir(os.path.join(dataset_folder, item)) else "📄", item)
```

```
Dataset folder:
D:\AIML LAB

Files available:
--------------------------------------------------
📁 iris
```

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Code

JupyterLab Python (ml env)

```
D:\AIML LAB

Files available:
--------------------------------------------------
iris
iris.zip
```

[8]:
```
import os

iris_folder = r"D:\AIML LAB\iris"

print("Iris dataset folder:")
print(iris_folder)

print("\nFiles inside the folder:")
print("-" * 60)

for item in os.listdir(iris_folder):
    full_path = os.path.join(iris_folder, item)
    print("" if os.path.isdir(full_path) else "", item)
```

```
Iris dataset folder:
D:\AIML LAB\iris

Files inside the folder:
------------------------------------------------------------
bezdekIris.data
Index
iris.data
iris.names
```

[9]:
```
import pandas as pd

file_path = r"D:\AIML LAB\iris\iris.data"


columns = [
    "sepal_length",
    "sepal_width",
    "petal_length",
    "petal_width",
    "species"
]

df = pd.read_csv(file_path, header=None, names=columns)

print("✓ Dataset loaded successfully!")
print(f"Rows    : {df.shape[0]}")
print(f"Columns : {df.shape[1]}")
```

```
✓ Dataset loaded successfully!
Rows    : 150
Columns : 5
```

[10]:
```
print("First 10 records")
print("-" * 80)

display(df.head(10))
```

```
First 10 records
--------------------------------------------------------------------------------
```

	sepal length	sepal width	petal length	petal width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
5	5.4	3.9	1.7	0.4	Iris-setosa
6	4.6	3.4	1.4	0.3	Iris-setosa
7	5.0	3.4	1.5	0.2	Iris-setosa
8	4.4	2.9	1.4	0.2	Iris-setosa

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Code JupyterLab Python (ml_env)

[11]:
```
print("Dataset Information")
print("-" * 80)

df.info()
```

```
Dataset Information
--------------------------------------------------------------------------------
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    str
dtypes: float64(4), str(1)
memory usage: 6.0 KB
```

[12]:
```
print("Statistical Summary")
print("-" * 80)

display(df.describe())
```

```
Statistical Summary
--------------------------------------------------------------------------------
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

[13]:
```
print("Missing Values")
print("-" * 80)

missing_values = df.isnull().sum()

display(missing_values.to_frame(name="Missing Count"))
```

```
Missing Values
--------------------------------------------------------------------------------
```

	Missing Count
sepal_length	0
sepal_width	0
petal_length	0
petal_width	0
species	0

[14]:
```
df_preprocessed = df.copy()

df_preprocessed.loc[0, "sepal_length"] = np.nan
df_preprocessed.loc[5, "sepal_width"] = np.nan
df_preprocessed.loc[10, "petal_width"] = np.nan

print("Missing values introduced for demonstration:")
print("-" * 60)

display(df_preprocessed.isnull().sum().to_frame(name="Missing Count"))
```

```
Missing values introduced for demonstration:
------------------------------------------------------------
```

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Code JupyterLab Python (ml_env)

[14]:

```
df_preprocessed = df.copy()

df_preprocessed.loc[0, "sepal_length"] = np.nan
df_preprocessed.loc[5, "sepal_width"] = np.nan
df_preprocessed.loc[10, "petal_width"] = np.nan

print("Missing values introduced for demonstration:")
print("-" * 60)

display(df_preprocessed.isnull().sum().to_frame(name="Missing Count"))
```

```
Missing values introduced for demonstration:
------------------------------------------------------------
```

	Missing Count
sepal length	1
sepal width	1
petal length	0
petal width	1
species	0

[15]:

```
numeric_columns = [
    "sepal_length",
    "sepal_width",
    "petal_length",
    "petal_width"
]

for column in numeric_columns:
    df_preprocessed[column] = df_preprocessed[column].fillna(
        df_preprocessed[column].mean()
    )

print("Missing values after mean imputation:")
print("-" * 60)

display(df_preprocessed.isnull().sum().to_frame(name="Missing Count"))
```

```
Missing values after mean imputation:
------------------------------------------------------------
```

	Missing Count
sepal length	0
sepal width	0
petal length	0
petal width	0
species	0

[16]:

```
print("Dataset after handling missing values")
print("-" * 80)

display(df_preprocessed.head(10))
```

```
Dataset after handling missing values
--------------------------------------------------------------------------------
```

	sepal length	sepal width	petal length	petal width	species
0	5.848322	3.500000	1.4	0.2	Iris-setosa
1	4.900000	3.000000	1.4	0.2	Iris-setosa
2	4.700000	3.200000	1.3	0.2	Iris-setosa
3	4.600000	3.100000	1.5	0.2	Iris-setosa
4	5.000000	3.600000	1.4	0.2	Iris-setosa
5	5.400000	3.048322	1.7	0.4	Iris-setosa

```
display(df_preprocessed.head(10))
```

```
Dataset after handling missing values
--------------------------------------------------------------------------------
```

	sepal length	sepal width	petal length	petal width	species
0	5.848322	3.500000	1.4	0.2	Iris-setosa
1	4.900000	3.000000	1.4	0.2	Iris-setosa
2	4.700000	3.200000	1.3	0.2	Iris-setosa
3	4.600000	3.100000	1.5	0.2	Iris-setosa
4	5.000000	3.600000	1.4	0.2	Iris-setosa
5	5.400000	3.048322	1.7	0.4	Iris-setosa
6	4.600000	3.400000	1.4	0.3	Iris-setosa
7	5.000000	3.400000	1.5	0.2	Iris-setosa
8	4.400000	2.900000	1.4	0.2	Iris-setosa
9	4.900000	3.100000	1.5	0.1	Iris-setosa

[17]:

```
df_preprocessed = df.copy()

print("✓ Preprocessing copy created successfully.")
print(f"Original dataset shape    : {df.shape}")
print(f"Preprocessing dataset shape: {df_preprocessed.shape}")
```

```
✓ Preprocessing copy created successfully.
Original dataset shape    : (150, 5)
Preprocessing dataset shape: (150, 5)
```

[18]:

```
df_preprocessed.loc[0, "sepal_length"] = np.nan
df_preprocessed.loc[5, "sepal_width"] = np.nan
df_preprocessed.loc[10, "petal_width"] = np.nan

print("Missing values introduced for demonstration")
print("-" * 60)

display(df_preprocessed.isnull().sum().to_frame(name="Missing Count"))
```

```
Missing values introduced for demonstration
------------------------------------------------------------
```

	Missing Count
sepal length	1
sepal width	1
petal length	0
petal width	1
species	0

[19]:

```
numeric_columns = [
    "sepal_length",
    "sepal_width",
    "petal_length",
    "petal_width"
]

for column in numeric_columns:
    df_preprocessed[column] = df_preprocessed[column].fillna(
        df_preprocessed[column].mean()
    )

print("✓ Missing values handled using Mean Imputation")
print("-" * 60)

display(df_preprocessed.isnull().sum().to_frame(name="Missing Count"))
```

```
✓ Missing values handled using Mean Imputation
```

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Code JupyterLab Python (ml env)

```
]

for column in numeric_columns:
    df_preprocessed[column] = df_preprocessed[column].fillna(
        df_preprocessed[column].mean()
    )

print("✓ Missing values handled using Mean Imputation")
print("-" * 60)

display(df_preprocessed.isnull().sum().to_frame(name="Missing Count"))
```

```
✓ Missing values handled using Mean Imputation
------------------------------------------------------------
```

	Missing Count
sepal length	0
sepal width	0
petal length	0
petal width	0
species	0

[20]:
```
print("Categorical Column")
print("-" * 60)

print("Column:", "species")
print("\nUnique categories:")

for category in df_preprocessed["species"].unique():
    print("•", category)
```

```
Categorical Column
------------------------------------------------------------
Column: species

Unique categories:
• Iris-setosa
• Iris-versicolor
• Iris-virginica
```

[21]:
```
from sklearn.preprocessing import MinMaxScaler


scaler = MinMaxScaler()

df_preprocessed[numeric_columns] = scaler.fit_transform(
    df_preprocessed[numeric_columns]
)

print("✓ Numerical features normalized using Min-Max Scaling")
print("-" * 60)
```

```
✓ Numerical features normalized using Min-Max Scaling
------------------------------------------------------------
```

[22]:
```
print("Dataset After Normalization")
print("-" * 80)

display(df_preprocessed.head(10))
```

```
Dataset After Normalization
--------------------------------------------------------------------------------
```

	sepal length	sepal width	petal length	petal width	species
0	0.430089	0.625000	0.067797	0.041667	Iris-setosa
1	0.166667	0.416667	0.067797	0.041667	Iris-setosa
2	0.111111	0.500000	0.050847	0.041667	Iris-setosa
3	0.083333	0.458333	0.084746	0.041667	Iris-setosa
4	0.194444	0.666667	0.067797	0.041667	Iris-setosa
5	0.305556	0.436801	0.118644	0.125000	Iris-setosa

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Code

JupyterLab Python (ml_env)

3	0.083333	0.458333	0.084746	0.041667	Iris-setosa
4	0.194444	0.666667	0.067797	0.041667	Iris-setosa
5	0.305556	0.436801	0.118644	0.125000	Iris-setosa
6	0.083333	0.583333	0.067797	0.083333	Iris-setosa
7	0.194444	0.583333	0.084746	0.041667	Iris-setosa
8	0.027778	0.375000	0.067797	0.041667	Iris-setosa
9	0.166667	0.458333	0.084746	0.000000	Iris-setosa

[23]:
```
print("Minimum Values After Normalization")
print("-" * 60)

display(df_preprocessed[numeric_columns].min().to_frame(name="Minimum"))

print("\nMaximum Values After Normalization")
print("-" * 60)

display(df_preprocessed[numeric_columns].max().to_frame(name="Maximum"))
```

```
Minimum Values After Normalization
------------------------------------------------------------
```

	Minimum
sepal length	0.0
sepal width	0.0
petal length	0.0
petal width	0.0

```
Maximum Values After Normalization
------------------------------------------------------------
```

	Maximum
sepal length	1.0
sepal width	1.0
petal length	1.0
petal width	1.0

[24]:
```
comparison = pd.DataFrame({
    "Feature": numeric_columns,
    "Original Mean": [
        df[column].mean() for column in numeric_columns
    ],
    "Normalized Mean": [
        df_preprocessed[column].mean() for column in numeric_columns
    ]
})

display(comparison)
```

	Feature	Original Mean	Normalized Mean
0	sepal length	5.843333	0.430089
1	sepal width	3.054000	0.436801
2	petal length	3.758667	0.467571
3	petal width	1.198667	0.460570

[25]:
```
print("FINAL PREPROCESSED DATASET")
print("-" * 80)

print("Rows    :", df_preprocessed.shape[0])
print("Columns :", df_preprocessed.shape[1])

print("\nMissing values:")
```

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Code JupyterLab Python (ml env)

```
print(df_preprocessed.isnull().sum().sum())

print("\n✓ Missing values handled")
print("✓ Categorical data encoded")
print("✓ Numerical data normalized")
```

```
FINAL PREPROCESSED DATASET
================================================================================
Rows    : 150
Columns : 5

Missing values:
0

✓ Missing values handled
✓ Categorical data encoded
✓ Numerical data normalized
```

[26]:
```
print("Final Preprocessed Dataset")
print("=" * 80)

display(df_preprocessed.head(15))
```

```
Final Preprocessed Dataset
================================================================================
```

	sepal length	sepal width	petal length	petal width	species
0	0.430089	0.625000	0.067797	0.041667	Iris-setasa
1	0.166667	0.416667	0.067797	0.041667	Iris-setasa
2	0.111111	0.500000	0.050847	0.041667	Iris-setasa
3	0.083333	0.458333	0.084746	0.041667	Iris-setasa
4	0.194444	0.666667	0.067797	0.041667	Iris-setasa
5	0.305556	0.436801	0.118644	0.125000	Iris-setasa
6	0.083333	0.583333	0.067797	0.083333	Iris-setasa
7	0.194444	0.583333	0.084746	0.041667	Iris-setasa
8	0.027778	0.375000	0.067797	0.041667	Iris-setasa
9	0.166667	0.458333	0.084746	0.000000	Iris-setasa
10	0.305556	0.708333	0.084746	0.460570	Iris-setasa
11	0.138889	0.583333	0.101695	0.041667	Iris-setasa
12	0.138889	0.416667	0.067797	0.000000	Iris-setasa
13	0.000000	0.416667	0.016949	0.000000	Iris-setasa
14	0.416667	0.833333	0.033898	0.041667	Iris-setasa

[1]:
```
import os

print("Current folder:", os.getcwd())
print("Notebook working directory files:")

for file in os.listdir():
    print(file)
```

```
Current folder: C:\Users\Lenovo\AIML SEM 5
Notebook working directory files:
.ipynb_checkpoints
Assignment 5 Matplotlib.ipynb
Assignment No. 6.ipynb
Experiment4.ipynb
Experiment5.ipynb
Seaborn Assignment 5.ipynb
Untitled.ipynb
```

[3]:
```
%cd "C:\Users\Lenovo\AIML SEM 5"

!git clone https://github.com/aaryanm1208-ship-it/AIML-LAB-EXPERIMENTS.git
```

```
C:\Users\Lenovo\AIML SEM 5
```

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Code JupyterLab Python (ml env)

```
aaryanm1208-ship-it
aaryanm1208@gmail.com
```

[6]:
```
%cd "C:\Users\Lenovo\AIML SEM 5\AIML-LAB-EXPERIMENTS"

!git add -- "Assignment No. 6.ipynb"
!git commit -m "Add Assignment 6 Iris Data Preprocessing"
```

```
C:\Users\Lenovo\AIML SEM 5\AIML-LAB-EXPERIMENTS
warning: in the working copy of 'Assignment No. 6.ipynb', LF will be replaced by CRLF the next time Git touches it
[main 39090c4] Add Assignment 6 Iris Data Preprocessing
 7 files changed, 1930 insertions(+), 765 deletions(-)
 delete mode 100644 Assignment 1/BFS AND DFS
 delete mode 100644 Assignment 4/On Jupyter/Set Up and Configure a Machine Learning Environment Using Anaconda and Jupiter Notebook, and Write a Basi
c Python Program to Perform Arithmetic and Statistical Calculations (e.g., mean, median, standard deviation)
 create mode 100644 Assignment No. 6.ipynb
 delete mode 100644 Assignment2/A* Algorithm
 delete mode 100644 Assignment3/TIC-TAC-TOE
 delete mode 100644 "Assignment5/Demonstrate the use of basic libraries in Python \342\200\242 Implementation of Python basic libraries such as Math,
Numpy and Scipy. \342\200\242 Perform Data Handling and Visualization Using Pandas, Matplotlib, Seaborn"
 delete mode 100644 README.md
```

[7]:
```
!git push origin main
```

```
To https://github.com/aaryanm1208-ship-it/AIML-LAB-EXPERIMENTS.git
   ce2d76d..39090c4  main -> main
```

[8]:
```
!git status
!git log -1 --oneline
```

```
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
39090c4 Add Assignment 6 Iris Data Preprocessing
```

[9]:
```
%cd "C:\Users\Lenovo\AIML SEM 5\AIML-LAB-EXPERIMENTS"
```

```
C:\Users\Lenovo\AIML SEM 5\AIML-LAB-EXPERIMENTS
```

[10]:
```
from pathlib import Path
import shutil

source_folder = Path(r"C:\Users\Lenovo\AIML SEM 5")
repo_folder = source_folder / "AIML-LAB-EXPERIMENTS"

copied = []

for file in source_folder.glob("*.ipynb"):
    if file.stem.strip().lower() in {
        "assignment no. 6",
        "assignment no 6",
        "assignment 6"
    }:
        print(f"Skipped: {file.name}")
        continue

    destination = repo_folder / file.name
    shutil.copy2(file, destination)
    copied.append(file.name)

print("\nCopied notebooks:")
for name in copied:
    print(name)

print(f"\nTotal copied: {len(copied)}")
```

```
Skipped: Assignment No. 6.ipynb

Copied notebooks:
Assignment 5 Matplotlib.ipynb
Experiment4.ipynb
Experiment5.ipynb
Seaborn Assignment 5.ipynb
Untitled.ipynb

Total copied: 5
```